Assignment 1
Agent-Based Adaptive Modelling
Modelling the Tumor–Immune Microenvironment

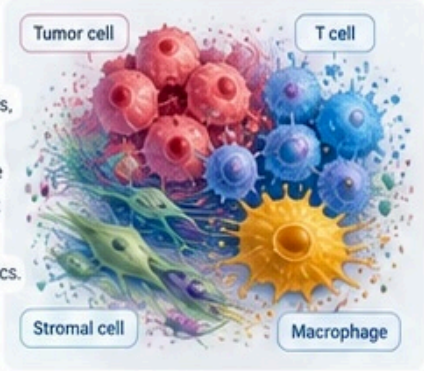
Course 1
Agent-Based Adaptive Modelling

Learning Goal

Develop a formal agent-based model of the tumor–immune microenvironment to analyse the **emergent dynamics of tumor growth and immune response.**

Context

The tumor microenvironment (TME) is a complex ecosystem where cancer cells, immune cells, stromal cells and soluble factors interact. The balance between tumor progression and immune control depends on local interactions, resource availability (e.g., nutrients and oxygen), and adaptive responses of the different cell populations. Agent-based modelling (ABM) allows us to explore how individual behaviours and local interactions can lead to emergent global dynamics.



Assignment Description

Students are asked to design and analyse an agent-based model of a simplified tumor microenvironment. The model should include at least the following components:

1. **Cell populations:** tumor cells, cytotoxic T cells, macrophages (at least three types).
2. **Environment:** spatial domain with nutrient and oxygen gradients.
3. **Behavioural rules:** cell movement, proliferation, death, immune activation, and local interactions (e.g., killing, inhibition, cytokine signalling).
4. **State variables and observables:** cell states, position, nutrient/oxygen levels, population counts.
5. **Hypotheses and experiments:** formulate at least two research questions and design simulation experiments to test them (e.g., effect of immune cell density, nutrient availability, or immune response strength on tumor growth).

Deliverables

- A written report (PDF, max 6 pages) including:
  - Model specification (agents, states, rules, environment)
  - Assumptions and simplifications (with biological justification)
  - Hypotheses and experimental design
  - Results and analysis (figures and discussion)
- A short presentation (10 minutes) summarizing the model, experiments and conclusions.

Duration
75–90 minutes

Group Size
3–4 students

Expected Tools
Any ABM framework or language (e.g., NetLogo, Python, Mesa, etc.)

Tip
You can choose to start from a simple model (e.g., 2D lattice) and progressively add complexity (e.g., different cell phenotypes, signalling factors, or spatial heterogeneity).